Monitoring quality drift

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete rate, slice and coverage record.

Keep sampling protocol, counts, denominator, threshold, slice mix, versions and unresolved causes together.

Keep one habit: Recall → W1–W3/W5 → independent denominator case → monitoring project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Monitoring compares current outcomes with a declared reference and sampling protocol. A change in observed quality may reflect case mix or instrumentation.

- For each window report eligible counts, passes, missing outcomes and condition mix. Alert when success falls below the toy **80%** threshold; investigate before attributing cause.

Release and rollback

> **My remembered idea, example and correction**

Queues and asynchronous work

> **My remembered idea, example and correction**

Sandboxing and resource limits

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Monitoring compares outcomes under a declared reference, sampling protocol and rubric. Main complete windows have reference 18/20 = 90% and current 14/20 = 70%, no missing outcomes: down 20 percentage points. Alert strictly below the fictional 80% threshold; equality does not alert. Record eligible counts, passes, failures, missing outcomes and condition mix.

  Added code protocol reports passes/(passes + observed failures) explicitly; missing outcomes raise a coverage alert and suspend full-window quality decision. Empty slices have unavailable rates, not 0% or 100%. Separate picture clear slice A/B is 18/20 = 90%, 19/20 = 95%; ambiguous is 8/10 = 80%, 4/10 = 40%. Totals 26/30 ≈ 86.7%, 23/30 ≈ 76.7%, down 10 percentage points.

  If B has no ambiguous questions, report that coverage gap; its clear-only 19/20 total has different case mix. Preserve rubric, version and instrumentation evidence. Small supplied samples show descriptive outcomes, not a stable trend, statistical significance or proven model-caused drift. Code uses constructed counts only.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Monitoring compares outcomes under a declared reference, sampling protocol and rubric. Main complete windows have reference 18/20 = 90% and current 14/20 = 70%, no missing outcomes: down 20 percentage points. Alert strictly below the fictional 80% threshold; equality does not alert. Record eligible counts, passes, failures, missing outcomes and condition mix.

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  If B has no ambiguous questions, report that coverage gap; its clear-only 19/20 total has different case mix. Preserve rubric, version and instrumentation evidence. Small supplied samples show descriptive outcomes, not a stable trend, statistical significance or proven model-caused drift. Code uses constructed counts only.

- Reference window passes 18/20; new window 14/20. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Monitoring compares outcomes under a declared reference, sampling protocol and rubric. Main complete windows have reference 18/20 = 90% and current 14/20 = 70%, no missing outcomes: down 20 percentage points. Alert strictly below the fictional 80% threshold; equality does not alert. Record eligible counts, passes, failures, missing outcomes and condition mix.

  Added code protocol reports passes/(passes + observed failures) explicitly; missing outcomes raise a coverage alert and suspend full-window quality decision. Empty slices have unavailable rates, not 0% or 100%. Separate picture clear slice A/B is 18/20 = 90%, 19/20 = 95%; ambiguous is 8/10 = 80%, 4/10 = 40%. Totals 26/30 ≈ 86.7%, 23/30 ≈ 76.7%, down 10 percentage points.

  If B has no ambiguous questions, report that coverage gap; its clear-only 19/20 total has different case mix. Preserve rubric, version and instrumentation evidence. Small supplied samples show descriptive outcomes, not a stable trend, statistical significance or proven model-caused drift. Code uses constructed counts only.

- A report omits this qualification: An alert identifies a condition for investigation, not a complete causal explanation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Monitoring compares outcomes under a declared reference, sampling protocol and rubric. Main complete windows have reference 18/20 = 90% and current 14/20 = 70%, no missing outcomes: down 20 percentage points. Alert strictly below the fictional 80% threshold; equality does not alert. Record eligible counts, passes, failures, missing outcomes and condition mix.

  Added code protocol reports passes/(passes + observed failures) explicitly; missing outcomes raise a coverage alert and suspend full-window quality decision. Empty slices have unavailable rates, not 0% or 100%. Separate picture clear slice A/B is 18/20 = 90%, 19/20 = 95%; ambiguous is 8/10 = 80%, 4/10 = 40%. Totals 26/30 ≈ 86.7%, 23/30 ≈ 76.7%, down 10 percentage points.

  If B has no ambiguous questions, report that coverage gap; its clear-only 19/20 total has different case mix. Preserve rubric, version and instrumentation evidence. Small supplied samples show descriptive outcomes, not a stable trend, statistical significance or proven model-caused drift. Code uses constructed counts only.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled monitoring quality **drift evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of monitoring quality **drift**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Monitoring quality **drift**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Monitoring quality **drift**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.
- A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.
- Compute both clear and ambiguous slice rates for A and B. Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

AI ACADEMY / AI-6/25

Quality slices: retain denominators

	Clear correct	Clear total	Ambiguous correct	Ambiguous total
Week A	18	20	8	10
Week B	19	20	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/25

Monitoring quality drift

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Clear rises 90% -> 95%; ambiguous falls 80% -> 40%.



2. Compute totals

A is 26/30; B is 23/30 correct.



3. Investigate ambiguity

Inspect that slice's failures before selecting a repair.

CHANGE ONE THING

Week B has no ambiguous questions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.
- Which labelled step matches this detail? Inspect that slice's failures before selecting a repair.
- 1. Compute totals
- 2. Investigate ambiguity
- 3. Compute slice rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Investigate ambiguity Inspect that slice's failures before selecting a repair. Slice monitoring exposes changes hidden by common easy questions.

Retain each denominator and compare the measurement conditions.

Changed-case question and optional hint

- Changed-case practice: Week B has no ambiguous questions. Which conclusion is supported by the supplied facts?
- 1. That slice is unmeasured, not perfectly correct.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That slice is unmeasured, not perfectly correct. No **observation**s means no rate can be calculated for it.

Repair and retain the record

Review Release and rollback (ai-6/24). Goal: Explain the **mechanism** and terminology of release and rollback. Short practice: Compute A's and B's correctness percentages.

Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?. Return to this week and explain what you changed.

Review Queues and asynchronous work (ai-6/22). Goal: Explain the **mechanism** and terminology of queues and asynchronous work. Short practice: Explain what the queued J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Monitoring quality drift: the case

AI ACADEMY / AI-6/25 · 01 · LOOK AT THE FACTS

Monitoring quality drift: the case

The situation

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.

Supplied evidence

A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

Explain monitoring quality drift using the deciding step in this fictional service.

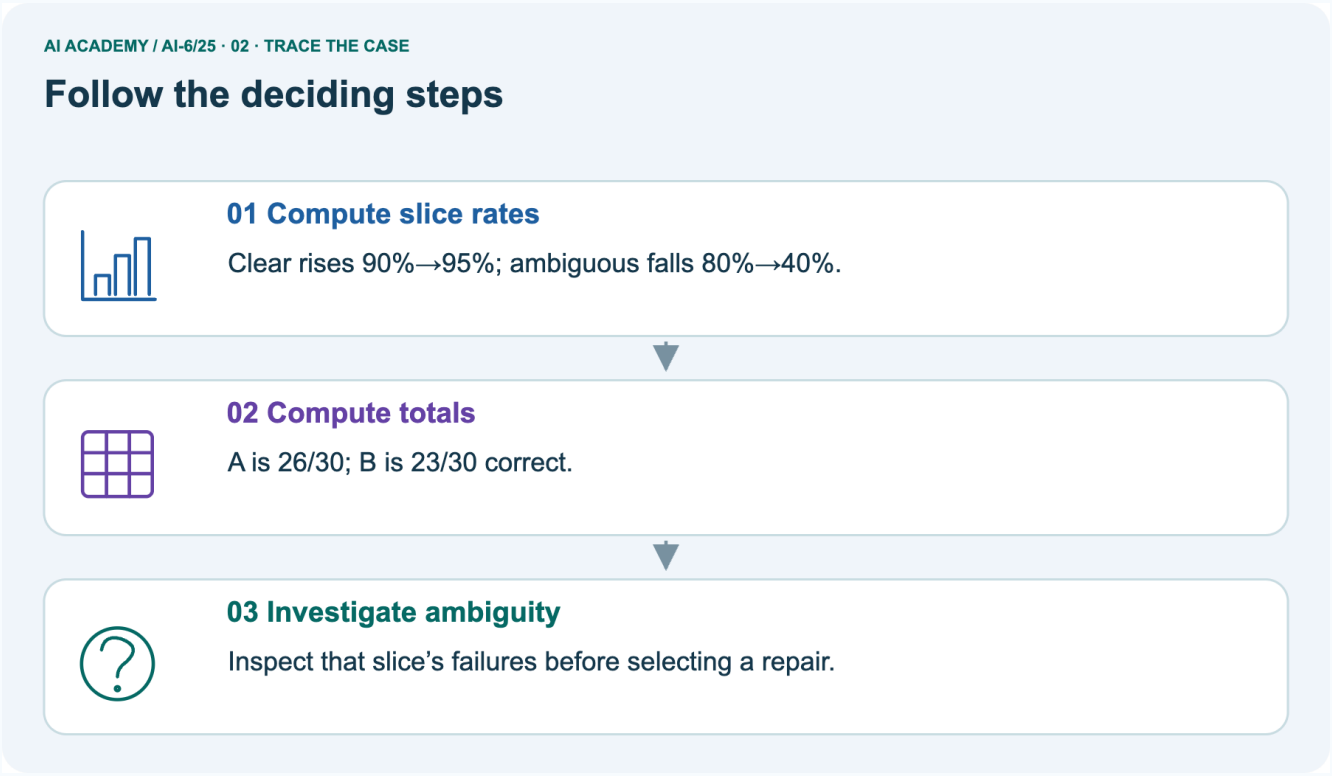
Follow the labelled picture with the matching explanation. Open the vector diagram

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks. Supplied facts: A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



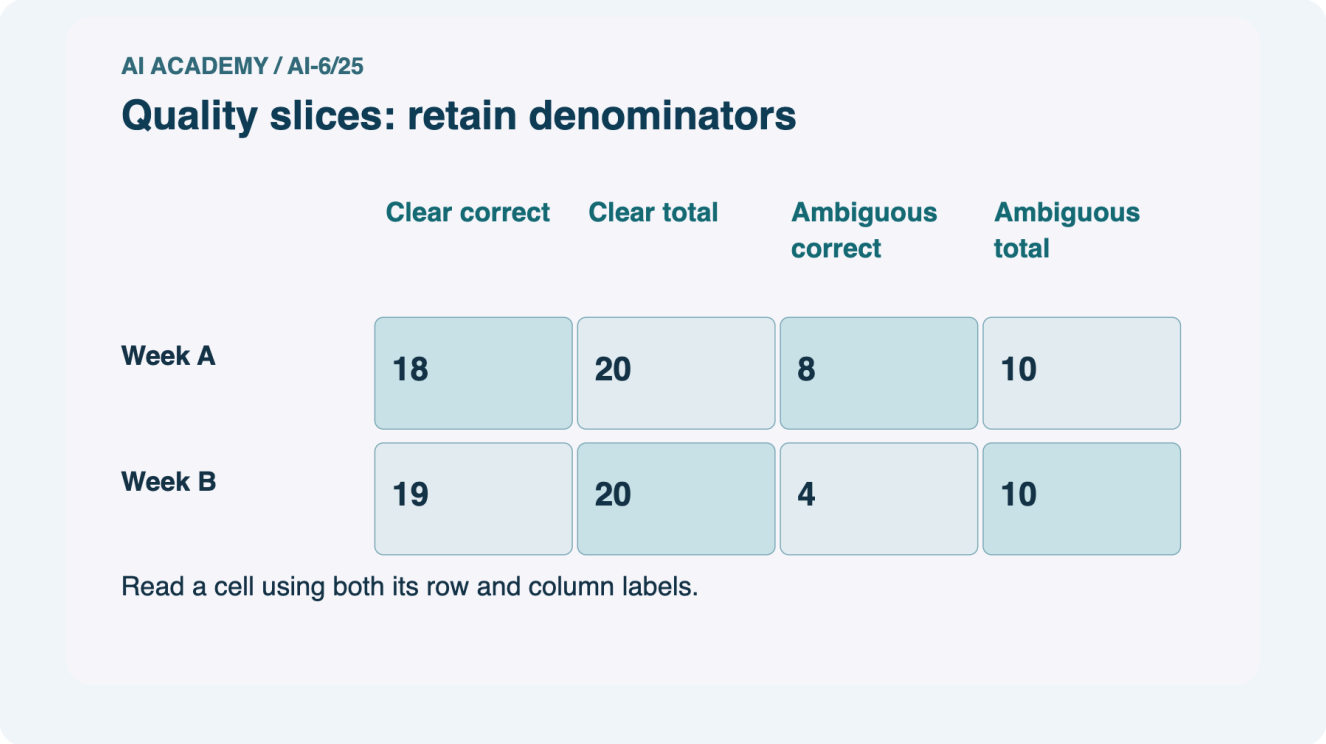
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Clear rises 90%→95%; ambiguous falls **80%**→40%. → Compute totals: A is 26/30; B is 23/30 correct. → Investigate ambiguity: Inspect that slice's failures before selecting a repair.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Quality slices: retain denominators



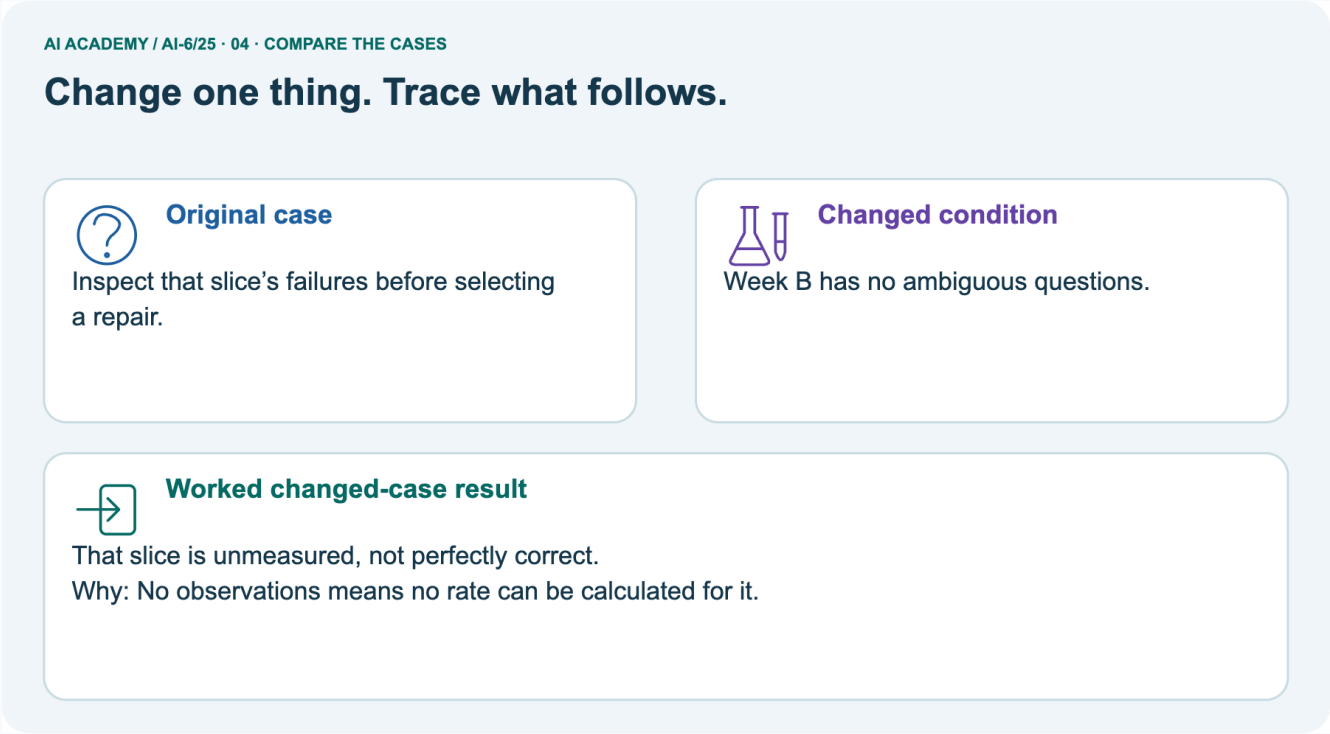
Follow the labelled picture with the matching explanation. Open the vector diagram

Quality slices: retain denominators. Week A: Clear correct = 18, Clear total = 20, Ambiguous correct = 8, Ambiguous total = 10. Week B: Clear correct = 19, Clear total = 20, Ambiguous correct = 4, Ambiguous total = 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

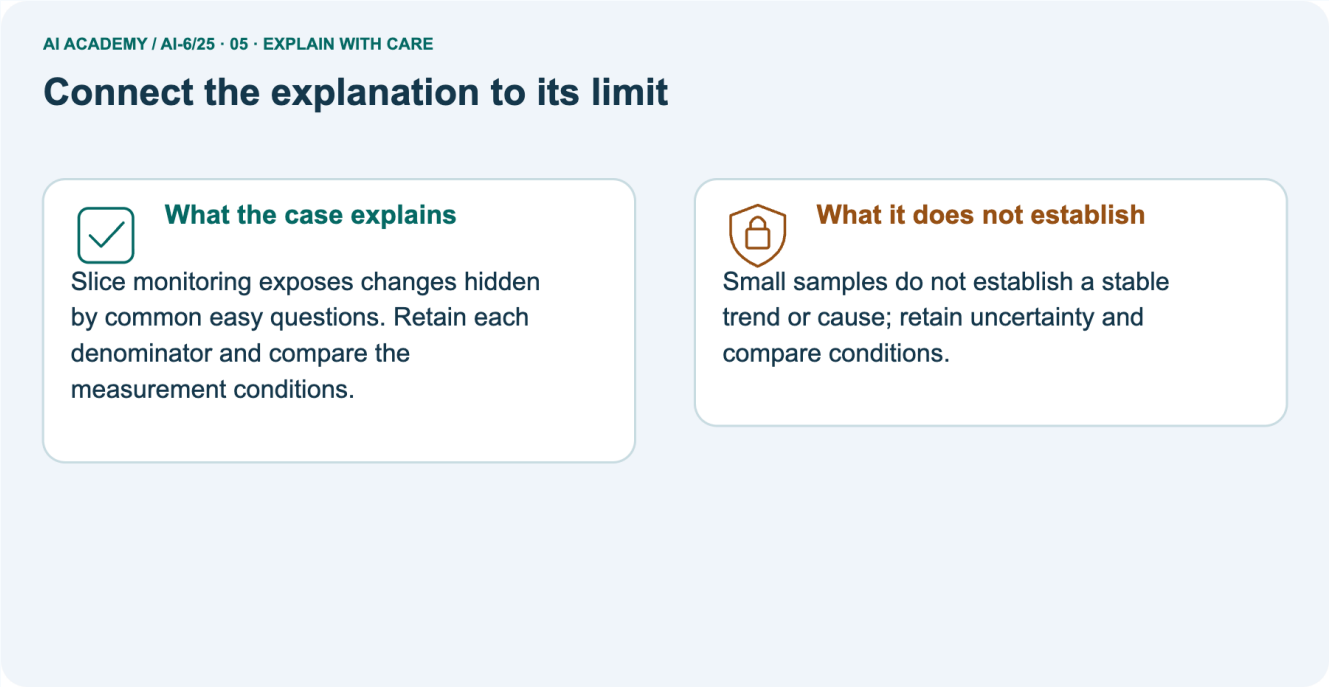
Original case: Inspect that slice's failures before selecting a repair. Changed condition: Week B has no ambiguous questions. Expected result: That slice is unmeasured, not perfectly correct.

Reason: No **observation**s means no rate can be calculated for it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

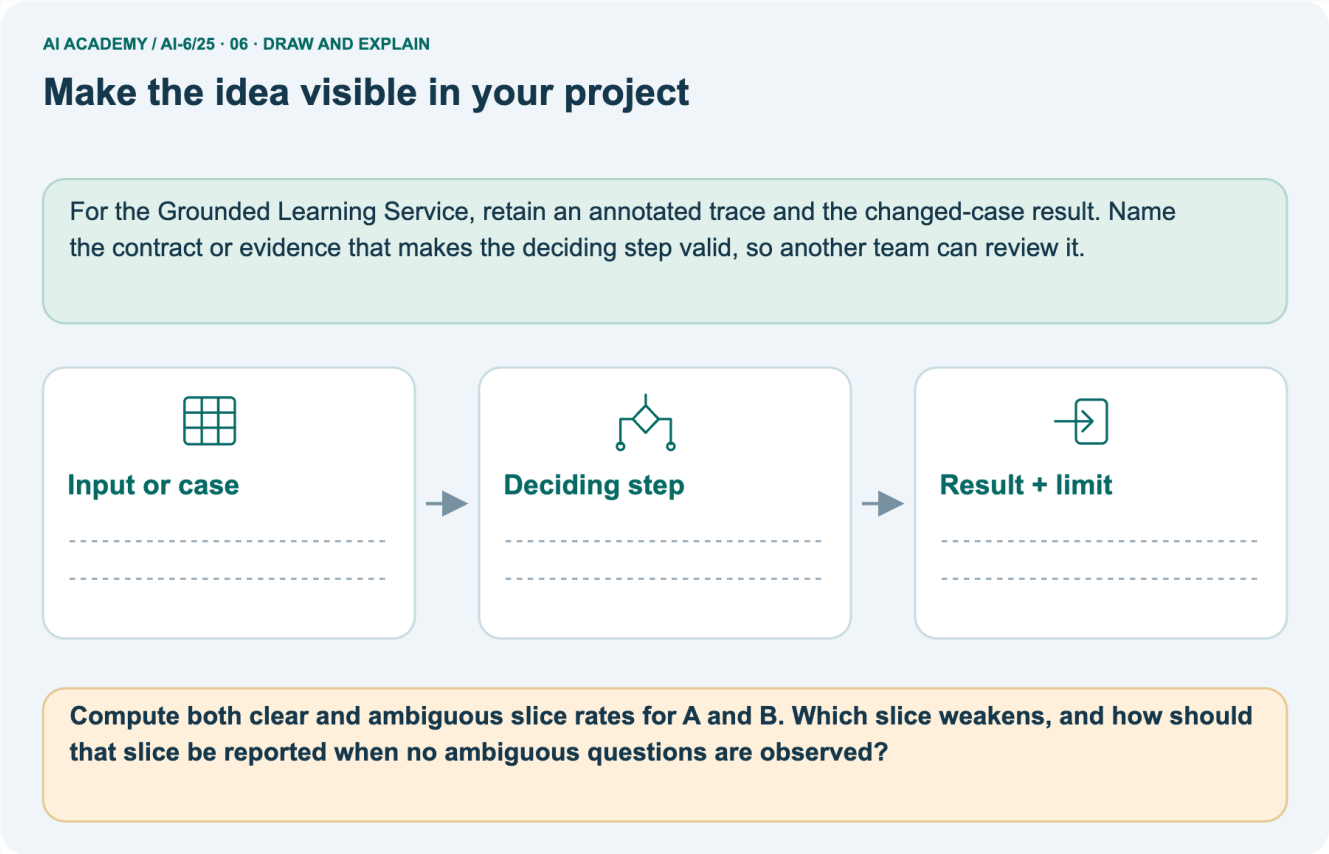
Slice monitoring exposes changes hidden by common easy questions. Retain each denominator and compare the measurement conditions.

Limit: Small samples do not establish a stable trend or cause; retain uncertainty and compare conditions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Monitoring quality drift: the case

02. Follow the deciding steps

03. Quality slices: retain denominators

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.